

# Elliptic Curves Over Finite Fields and Applications to Cryptography

2026 School on Arithmetic, Geometry and Algorithms

Juanita Duque-Rosero

Bernoulli Institute, University of Groningen.

[j.duque.rosero@rug.nl](mailto:j.duque.rosero@rug.nl)

# Introduction

These lecture notes accompany the lectures for the course on elliptic curves over finite fields and applications to cryptography for the 2026 School on Arithmetic, Geometry and Algorithms. The main references I used are [Kun26, Sil09]. Other useful references are [HPS14, AM26].

At the heart of public-key cryptography are functions that are easy to compute but are believed hard to reverse without secret information. In these notes, we first build the arithmetic of elliptic curves over finite fields from explicit formulas. We then use it to formulate the discrete logarithm problem and elliptic-curve Diffie–Hellman. At the end, we will discuss isogenies and their cryptographic context.

## Explicit computations

The ideal way to follow these notes is to try examples in your favorite computer algebra system. I recommend you use the open source software [SageMath](#), but you are welcome to use whatever you prefer ([Magma](#), [Oscar](#), [PARI/GP](#), etc.).

## Prerequisites

The required background includes arithmetic in  $\mathbb{F}_p$ : modular addition, multiplication, inverses, and polynomial evaluation. Basic group theory is also used in the course.

## Road map

Lectures 1–2 form the core course: points, the group law, point counts, ECDLP, and ECDH. Lecture 3 introduces isogenies and supersingular isogeny graphs. Lectures 4–5 are optional advanced material on endomorphism rings, quaternion algebras, and computational problems.

## A note from the author

Despite my best efforts, these notes may contain typos. If you spot any, please feel free to email me; I appreciate your help!

## Acknowledgments

These notes were inspired by the Arizona Winter School 2026: Computational Aspects of Arithmetic Geometry and Cryptography, and the Preliminary Arizona Winter School 2025: Mathematical cryptography and algorithms in number theory. Thank you to the speakers, Sarah Arpin, Kirsten Eisentraeger, David Harvey, Sabrina Kunzweiler, Chloe Martindale,

and John Voight and to the organizers Renee Bell, Daniel Erman, Brandon Levin, Anna Medvedovsky, Padma Srinivasan, Anthony Várilly-Alvarado, Isabel Vogt, and Hang Xue.

# Contents

|          |                                           |          |
|----------|-------------------------------------------|----------|
| <b>1</b> | <b>Elliptic curves over finite fields</b> | <b>1</b> |
| 1.1      | A short toolkit                           | 1        |
| 1.1.1    | Arithmetic in $\mathbb{F}_p$              | 1        |
| 1.2      | Curves and their points                   | 2        |
| 1.2.1    | Points on elliptic curves                 | 3        |
| 1.2.2    | Counting points by hand                   | 4        |
| 1.3      | The group law                             | 5        |
| 1.4      | Isomorphisms and the $j$ -invariant       | 7        |
| 1.5      | How many points can there be?             | 7        |
| 1.6      | Practice problems                         | 8        |

# Lecture 1: Elliptic curves over finite fields

This course starts with arithmetic in prime fields. Our immediate goal is to construct a finite abelian group from a polynomial equation and learn to compute in it. The cryptographic applications in later lectures use exactly this arithmetic, but with primes and subgroup orders too large to enumerate.

Throughout this lecture,  $p > 3$  is a prime.

## 1.1 A short toolkit

### 1.1.1 Arithmetic in $\mathbb{F}_p$

The elements of  $\mathbb{F}_p$  are  $0, 1, \dots, p-1$ , with addition and multiplication reduced modulo  $p$ . Every nonzero element  $a \in \mathbb{F}_p$  has a unique inverse  $a^{-1}$ , so division by  $a$  means multiplication by  $a^{-1}$ . For example, in  $\mathbb{F}_7$  we have  $2^{-1} = 4$ , since  $2 \cdot 4 = 8 = 1$ .

**Definition 1.1.** An element  $r \in \mathbb{F}_p$  is a **quadratic residue** if  $r = y^2$  for some  $y \in \mathbb{F}_p$ . Otherwise it is a **quadratic nonresidue**.

For  $p$  odd, every nonzero quadratic residue has exactly two square roots,  $y$  and  $-y$ , while 0 has one. Thus a table of squares is enough to count points on the curves in this lecture.

**Example 1.2.** The squares in  $\mathbb{F}_7$  are

$$0^2 = 0, \quad 1^2 = 6^2 = 1, \quad 2^2 = 5^2 = 4, \quad 3^2 = 4^2 = 2.$$

Hence the quadratic residues in  $\mathbb{F}_7$  are 0, 1, 2, 4.

 **Exercise 1.3.** In this problem, you will study quadratic residues in finite fields.

1. Find the quadratic residues in  $\mathbb{F}_{11}$  and  $\mathbb{F}_{13}$ . How many are there in each case?
2. Can you find a pattern in the number of quadratic residues in  $\mathbb{F}_p$  for odd primes  $p$ ?
3. Can you prove the formula you found? *Hint: consider the map  $x \mapsto x^2$  on the nonzero elements of  $\mathbb{F}_p$ .*

## 1.2 Curves and their points

An elliptic curve over a field  $k$  is a geometric object described here by an equation in two variables. In this course we use the **short Weierstrass equation**

$$y^2 = x^3 + ax + b,$$

where  $a$  and  $b$  are chosen constants in  $k$ . Usually,  $k$  will denote the finite fields  $\mathbb{F}_p$  or  $\mathbb{F}_{p^2}$ . A solution  $(x, y) \in k^2$  is called an **affine point** of the curve, or simply a point of the curve.

For example, on  $y^2 = x^3 + x + 1$  over  $\mathbb{F}_7$ , taking  $x = 2$  gives

$$y^2 = 2^3 + 2 + 1 = 4.$$

In 1.2, we saw that 4 is a quadratic residue in  $\mathbb{F}_7$ , so there are two values of  $y$  that satisfy the equation. Since  $2^2 = 4$  and  $5^2 = 4$  in  $\mathbb{F}_7$ , this value  $x = 2$  gives the two affine points  $(2, 2)$  and  $(2, 5)$ . For a fixed  $x$ , there can similarly be zero, one, or two affine points, according as the right-hand side is a nonresidue, zero, or a nonzero quadratic residue.

Finally, not every equation of this shape gives an elliptic curve. We require that it have no singular point—no crossing or cusp in its geometric picture. For a short Weierstrass equation in characteristic different from 2 and 3, this is exactly the nonvanishing condition in the following definition. It is what makes the addition law well behaved.

**Definition 1.4.** An elliptic curve in short Weierstrass form over  $\mathbb{F}_p$  is an equation

$$E: y^2 = x^3 + ax + b, \quad a, b \in \mathbb{F}_p, \quad (1.5)$$

such that

$$4a^3 + 27b^2 \neq 0.$$

Its set of  $\mathbb{F}_p$ -points is

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 = x^3 + ax + b\} \cup \{O\}.$$

The extra symbol  $O$ , called the **point at infinity**, will be the identity element of the group law.

To see where  $O$  comes from, homogenize the equation:

$$Y^2Z = X^3 + aXZ^2 + bZ^3.$$

In projective coordinates the solutions with  $Z = 0$  consist of the single point  $[0 : 1 : 0]$ , represented in the affine formulas by  $O$ . Including this point is what allows point addition to have an identity.

*Remark 1.6.* The restriction  $p > 3$  is deliberate: it makes this short equation and the formulas below valid without exceptional characteristic-2 or characteristic-3 cases.

 **Exercise 1.7.** Graph the real solutions of  $y^2 = f(x)$  for

$$f(x) = x^3 - 1, \quad f(x) = x^3 - x, \quad f(x) = x^3 - x^2.$$

Which of these equations are nonsingular and hence define elliptic curves over  $\mathbb{R}$ ?

**Definition 1.8.** The discriminant of (1.5) is

$$\Delta(E) = -16(4a^3 + 27b^2).$$

The condition  $\Delta(E) \neq 0$  says that the cubic on the right has no repeated root. In this case, we say that  $E$  is nonsingular. If  $\Delta(E) = 0$ , then  $E$  is singular and has a cusp or crossing point.

 **Exercise 1.9.** For which primes  $p > 3$  does  $y^2 = x^3 + x + 2$  fail to define an elliptic curve over  $\mathbb{F}_p$ ? What about over  $\mathbb{F}_{p^2}$ ?

 **Exercise 1.10.** Let  $k$  be a field of characteristic different from 2 and 3, and let  $f(x) = x^3 + ax + b$ .

1. Show that  $4a^3 + 27b^2 = 0$  if and only if  $f$  has a repeated root.
2. Use the partial derivatives of  $g(x, y) = y^2 - x^3 - ax - b$  to show that  $f$  having a repeated root is equivalent to the existence of a singular affine point on the curve.

### 1.2.1 Points on elliptic curves

The same equation can be studied over any field containing its coefficients.

**Definition 1.11.** Let  $k$  and  $K$  be fields with  $k \subseteq K$  and  $k$  being a subfield of  $K$ . Then we say that  $K$  is an extension field of  $k$ . We denote this by  $K/k$ .

If  $E$  is defined over  $k$  and  $K/k$  is a field extension, then

$$E(K) := \{(x, y) \in K^2 : y^2 = x^3 + ax + b\} \cup \{O\}.$$

The notation  $\#E(K)$  denotes the number of these points when the set is finite.

**Example 1.12** (One curve over several fields). Consider the nonsingular short Weierstrass equation

$$E: y^2 = x^3 - 27x + 2970,$$

Over  $\mathbb{C}$ , every value of  $x$  except a root of the cubic gives two values of  $y$ , so  $E(\mathbb{C})$  is infinite. Over  $\mathbb{R}$ ,

$$x^3 - 27x + 2970 = (x + 15)(x^2 - 15x + 198),$$

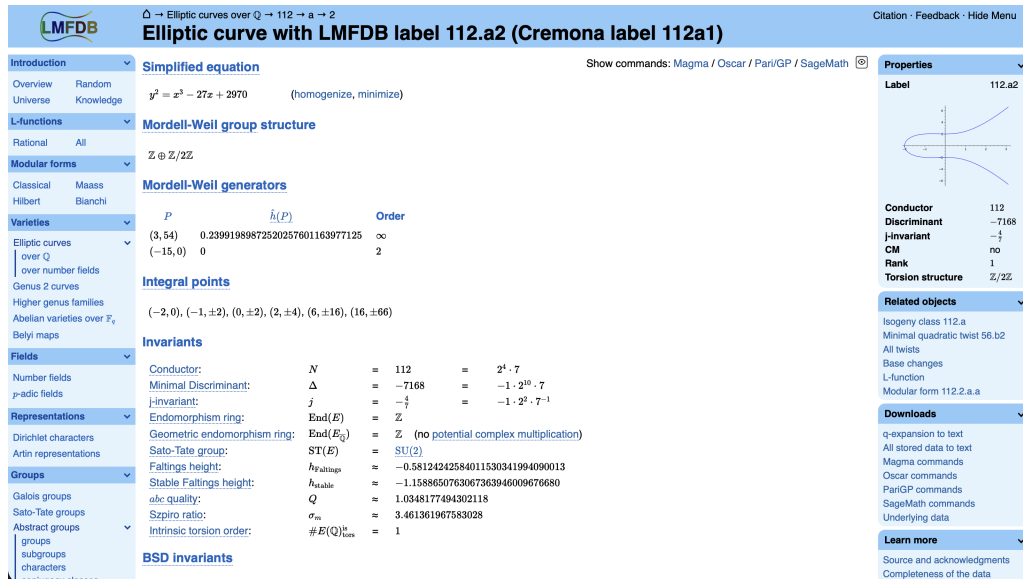
and  $x^2 - 15x + 198 > 0$  for every real  $x$ . Thus  $E(\mathbb{R})$  has affine points exactly when  $x \geq -15$ , including  $(-15, 0)$ . Over  $\mathbb{Q}$ , the points  $(-15, 0)$ ,  $(3, \pm 54)$ , and  $(21, \pm 108)$  are rational.

Over  $\mathbb{F}_{11}$ , direct enumeration gives

$$E(\mathbb{F}_{11}) = \{O, (0, 0), (2, \pm 3), (3, \pm 1), (4, 0), (5, \pm 1), (7, 0), (10, \pm 2)\},$$

so  $\#E(\mathbb{F}_{11}) = 12$ . Over the quadratic extension  $\mathbb{F}_{11^2}$  the same curve has 144 points. The points over  $\mathbb{F}_{11}$  are included among these extension-field points. Questions about integral points, obtained by asking for integer coordinates rather than field-valued coordinates, lead to deeper arithmetic geometry; see [Zag87].

A useful source for further examples and arithmetic data is the [LMFDB elliptic-curve database](#). The page for the curve in Example 1.12 is [this](#). It contains a wealth of information, including the number of points over many finite fields, the group structure of  $E(\mathbb{F}_p)$  for many primes  $p$ , and the torsion subgroup over  $\mathbb{Q}$ .



**Exercise 1.13.** For each curve below, compute  $\#E(\mathbb{F}_5)$ . Then use SageMath to compute  $\#E(\mathbb{F}_{5^2})$ . Do you see a pattern?

$$y^2 = x^3 - x, \quad y^2 = x^3 + x + 1.$$

### 1.2.2 Counting points by hand

For each  $x \in \mathbb{F}_p$ , compute  $x^3 + ax + b$  and look it up in the table of quadratic residues. A nonzero residue gives two points, 0 gives one, and a nonresidue gives none. Finally add  $O$ .

**Example 1.14.** Let  $E/\mathbb{F}_7$  be

$$E: y^2 = x^3 + x + 1.$$



The values of  $x^3 + x + 1$  for  $x = 0, 1, \dots, 6$  are

$$1, 3, 4, 3, 6, 5, 6.$$

Only 1 and 4 are quadratic residues in this list. Therefore

$$E(\mathbb{F}_7) = \{O, (0, 1), (0, 6), (2, 2), (2, 5)\}, \quad \#E(\mathbb{F}_7) = 5.$$

 **Exercise 1.15.** Count the points on each curve over  $\mathbb{F}_7$ :

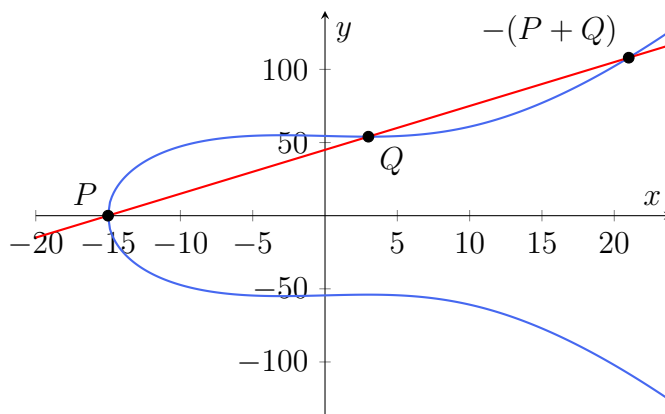
$$y^2 = x^3 - x, \quad y^2 = x^3 + 3.$$

Check that their point counts are 8 and 13, respectively.

## 1.3 The group law

Over the real numbers one can picture the group law by drawing a line through two points, finding the third intersection, and reflecting in the  $x$ -axis. Over  $\mathbb{F}_p$  there is no picture, but the same algebraic formulas work using modular inverses.

Equivalently, three collinear points on a nonsingular cubic add to  $O$  (with intersection multiplicities counted). For the real curve in Example 1.12, the line through  $P = (-15, 0)$  and  $Q = (3, 54)$  also meets the curve at  $R = (21, 108)$ . Thus  $P + Q + R = O$ , and  $P + Q = (21, -108)$ .



**Theorem 1.16** (Addition formulas). *Let  $E : y^2 = x^3 + ax + b$  be an elliptic curve over  $\mathbb{F}_p$ .*

1.  *$O$  is the identity:  $P + O = O + P = P$ .*
2. *The inverse of  $(x, y)$  is  $-(x, y) = (x, -y)$ .*
3. *If  $P = (x_1, y_1)$  and  $Q = (x_2, y_2)$  with  $x_1 \neq x_2$ , put*

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_3 = m^2 - x_1 - x_2, \quad y_3 = m(x_1 - x_3) - y_1.$$

*Then  $P + Q = (x_3, y_3)$ .*

4. *If  $P = (x_1, y_1)$  with  $y_1 \neq 0$ , put*

$$m = \frac{3x_1^2 + a}{2y_1}, \quad x_3 = m^2 - 2x_1, \quad y_3 = m(x_1 - x_3) - y_1.$$

*Then  $[2]P = (x_3, y_3)$ .*

5. *If  $Q = -P$ , including the case  $y_1 = y_2 = 0$ , then  $P + Q = O$ .*

*With these rules,  $(E(\mathbb{F}_p), +)$  is a finite abelian group.*

The associativity assertion is the hard part of this theorem; it follows from the projective geometry of a nonsingular cubic. A proof of the group law and the formulas appears in [Sil09, Ch. III, §§2–3]. We will use the theorem, not attempt that proof here.

**Example 1.17.** Continue with Example 1.14. For  $P = (0, 1)$ , doubling gives

$$m = \frac{1}{2} = 4 \pmod{7}, \quad [2]P = (2, 5).$$

Adding  $P$  to  $(2, 5)$  gives  $[3]P = (2, 2)$ , then  $[4]P = (0, 6)$ , and finally  $[5]P = O$ . Thus  $P$  generates all of  $E(\mathbb{F}_7)$ .

**Definition 1.18.** The scalar multiplication  $[n]P$  is computed efficiently by double-and-add: write  $n$  in binary, repeatedly double the current point, and add only at the binary digits equal to 1. This needs only about  $\log_2 n$  doublings and additions.

 **Exercise 1.19.** Use double-and-add to compute  $[13]P$  for  $P = (0, 1)$  on the curve in Example 1.14. Explain why the answer could also have been found by reducing 13 modulo  $\text{ord}(P)$ .

## 1.4 Isomorphisms and the $j$ -invariant

**Definition 1.20.** Let  $E : y^2 = x^3 + ax + b$  and  $E' : y^2 = x^3 + a'x + b'$  be short Weierstrass curves over  $\mathbb{F}_p$ . They are **isomorphic over  $\mathbb{F}_p$**  if there is a  $u \in \mathbb{F}_p^\times$  such that

$$a' = u^4a, \quad b' = u^6b.$$

The isomorphism sends  $(x, y)$  to  $(u^2x, u^3y)$  and sends  $O$  to  $O$ .

**Definition 1.21.** The  $j$ -invariant of  $E$  is

$$j(E) = 1728 \frac{4a^3}{4a^3 + 27b^2} = \frac{256 \cdot 27a^3}{4a^3 + 27b^2}.$$

It is unchanged by isomorphism. Over an algebraic closure, two elliptic curves of characteristic different from 2 and 3 are isomorphic exactly when their  $j$ -invariants agree.

**Example 1.22.** Over  $\mathbb{F}_7$ , the curves  $y^2 = x^3 + x$  and  $y^2 = x^3 + 2x$  are isomorphic: take  $u = 2$ , for which  $u^4 = 2$  in  $\mathbb{F}_7$ . The map is  $(x, y) \mapsto (4x, y)$ .

 **Exercise 1.23.** Verify directly that  $(x, y) \mapsto (4x, y)$  sends points of  $y^2 = x^3 + x$  over  $\mathbb{F}_7$  to points of  $y^2 = x^3 + 2x$ . Find its inverse map, and compute the common  $j$ -invariant in  $\mathbb{F}_7$ .

## 1.5 How many points can there be?

The elementary bound  $\#E(\mathbb{F}_q) \leq 2q + 1$  follows by counting the at most two  $y$ -values above each  $x$ . Hasse's theorem is vastly sharper; see [Sil09, Ch. V, §1].

**Theorem 1.24** (Hasse). *If  $E$  is an elliptic curve over  $\mathbb{F}_q$ , then*

$$q + 1 - 2\sqrt{q} \leq \#E(\mathbb{F}_q) \leq q + 1 + 2\sqrt{q}.$$

For  $q = 7$ , the possible integer values lie between 3 and 13. This explains why the examples above can have dramatically different point counts, even though both are defined over the same field.

**Example 1.25.** The two ends of Hasse's interval already occur over  $\mathbb{F}_7$ :

$$E_1 : y^2 = x^3 - 3 \quad \text{has } \#E_1(\mathbb{F}_7) = 3, \quad E_2 : y^2 = x^3 + 3 \quad \text{has } \#E_2(\mathbb{F}_7) = 13.$$

Thus the variation in point counts is real, even for a very small field.

 **Exercise 1.26.** Enumerate the curves  $y^2 = x^3 + ax + b$  over  $\mathbb{F}_7$  with nonzero discriminant. Verify that the curves in Example 1.25 attain the smallest and largest possible point counts.

 **Exercise 1.27.** For  $p = 5, 7, 11$ , use SageMath to enumerate all pairs  $(a, b)$  with  $4a^3 + 27b^2 \neq 0$  and record the possible values of  $\#E(\mathbb{F}_p)$ . Compare your data with Hasse's interval.

 **Exercise 1.28.** For several primes  $p$ , compute  $\lfloor p + 1 + 2\sqrt{p} \rfloor$  and enumerate all nonsingular curves  $y^2 = x^3 + ax + b$  over  $\mathbb{F}_p$ . Let  $M(p)$  be the number of pairs  $(a, b)$  whose curve has this maximal permitted number of points. Record the values of  $M(p)$  and look for a pattern.

## 1.6 Practice problems

The following problems collect the operations used throughout the lecture. Work the first three by hand before checking any computations with SageMath.

 **Exercise 1.29.** Over  $\mathbb{F}_{11}$ , consider

$$E: y^2 = x^3 + 2x + 7.$$

Compute the discriminant and verify that  $E$  is nonsingular. Make a table of the quadratic residues in  $\mathbb{F}_{11}$ , then use it to enumerate  $E(\mathbb{F}_{11})$ .

 **Exercise 1.30.** Use your point count for  $E/\mathbb{F}_{11}$  to check Hasse's bound. Determine the group structure of  $E(\mathbb{F}_{11})$ .

These are the same operations implemented by an elliptic-curve cryptographic library. The only difference in practice is the size of the field and the subgroup: production parameters make direct enumeration and trial search infeasible.

# Bibliography

- [AM26] Sarah Arpin and ChloeE Martindale. Arizona Winter School (paws) lectures, 2026. [swc-math.github.io](https://swc-math.github.io). ↑[ii](#).
- [HPS14] Jeffrey Hoffstein, Jill Pipher, and Joseph H. Silverman. *An introduction to mathematical cryptography*. Undergraduate Texts in Mathematics. Springer, New York, second edition, 2014. ↑[ii](#).
- [Kun26] Sabrina Kunzweiler. Preliminary Arizona Winter School (paws) lectures, 2026. [swc-math.github.io](https://swc-math.github.io). ↑[ii](#).
- [Sil09] Joseph H. Silverman. *The arithmetic of elliptic curves*, volume 106 of *Graduate Texts in Mathematics*. Springer, Dordrecht, second edition, 2009. ↑[ii](#), [6](#), [7](#).
- [Zag87] Don Zagier. Large integral points on elliptic curves. *Math. Comp.*, 48(177):425–436, 1987. ↑[4](#).